

User Manual

Reading the tool honestly.

Preselection guide for guide-vane bearings in reversible pump-turbines — elastic (deva.tex 552) vs. rigid (deva.bm 392).

PREFACE

This tool stands at the *beginning* of a conversation — never at its end.

It is an opener, not a final authority. It shows which damage mechanism plausibly governs a load situation and where an elastic bearing may outperform a rigid one — or the other way round. It does not predict service life, it does not compute exact forces away from its single calibration point, and it does not claim to replace machine data, FE or CFD. What it cannot represent, it does not pretend to represent: every number on screen carries its own evidence class.

The right outcome of a session with this tool is therefore not a material decision but a **better next question**: which measurement, which drawing, which operating data would settle the point that the tool made visible.

How to use this manual: Chapter 1 explains the toolbar, your permanent entry point. Chapter 2 explains how valid a number is. Chapter 3 goes through every section of the tool — what you see, what you can assess, what you cannot. Chapter 4 gives a short conversation workflow.

The toolbar — *your control centre*

On desktop the toolbar is a **sidebar on the right edge**; it stays in place while you scroll. Click the header strip (> > > QUICK CONTROLS) to collapse it to a thin strip and click again to expand it; the tool remembers your choice. On narrow desktop windows (below approx. 1100 px) it starts collapsed. On iPad and phone the same controls sit in a **bottom bar** with a compact jump row.

The image shows a software interface with two main panels. The left panel, titled 'QUICK CONTROLS', contains several sections: 'LIVE CONTROLS' with a heading 'Adjust the operating point' and a note 'Every change updates the charts instantly. Stays with you as you scroll.'; 'CYCLE ANIMATION' with play and stop buttons and a 'CLOSING' button; an 'EXTRAPOLATED' status bar; 'OPERATING MODE' with 'Pump' and 'Power Gen' buttons; and four sliders for 'GUIDE VANE ANGLE' (15°), 'RADIAL CLEARANCE Δd' (0.200 mm), 'SHOCK PULSE T' (5 ms), and 'Hard limits · tex ✓ · bm ✓'. The right panel, titled 'TOOL MAP · JUMP TO', lists ten tools with evidence class badges (A, B, C, D) and descriptions. At the bottom, it explains the evidence classes and provides a link to the 'User Manual (PDF)'.

QUICK CONTROLS

LIVE CONTROLS

Adjust the operating point

Every change updates the charts instantly.
Stays with you as you scroll.

CYCLE ANIMATION

▶ ◻ CLOSING

EXTRAPOLATED

Hard limits · tex ✓ · bm ✓

OPERATING MODE

Pump Power Gen

GUIDE VANE ANGLE 15°

0° closed 90° open

RADIAL CLEARANCE Δd 0.200 mm

0.05 0.20 ref 0.35

SHOCK PULSE T 5 ms

1 fast 50 quasi-static

TOOL MAP · JUMP TO

- A** **Validity window**
How far from the calibration anchor?
- C** **Bearing forces F1–F3**
Load split at the current angle
- C** **Force vs. angle**
Where in the closing sweep forces rise
- B/C** **Stress proxy**
FE-anchored peak vs. limit per load regime
- B** **Hard-limit screen**
p · pU · v vs. datasheet — exclusion test
- D** **Vibration spectrum**
Broad vs. selective response (synthetic)
- D** **Resonance position**
f₀ vs. excitation f_a (f₀ assumed)
- A/D** **Modal parameter set**
E' · tan δ · ζ · Q — what drives it
- D** **Shock response**
When compliance becomes protective
- C** **Snapshot report**
Current point, exportable
- C** **Cycle report**
Full 90° → 0° sweep, dominant mechanism

HOW VALID IS A NUMBER?

A MEASURED **B DERIVED**
C HEURISTIC **D HYPOTHESIS**

One calibration anchor: power · 0° · Δd 0.200 mm.
Everything else is reconstructed or extrapolated.
The tool opens a conversation — it does not close one.

[User Manual \(PDF\)](#)

- Cycle animation.** ▶ plays the closing sweep 90° → 0°; ↺ loops it. The pill shows the operating phase (idle · normal · transition · closing).
- Validity pill.** Live distance from the calibration anchor: **CALIBRATED ANCHOR** → **NEAR ANCHOR** → **EXTRAPOLATED** → **FAR EXTRAPOLATION**. Read every output through this pill.
- Hard-limit line.** tex ✓ · bm ✓ — does each material pass its datasheet limits (projected pressure, pU, sliding speed) at the current point?
- Operating mode.** Pump / Power generation. Only power mode is anchored — pump mode is always extrapolated.
- Guide-vane angle** (0° closed ... 90° open). Anchor at 0°.
- Radial clearance Δd** (0.05 ... 0.35 mm). Anchor at 0.200 mm. Definition (radial vs. diametral) still to be verified against source.
- Shock pulse τ** (1 ... 50 ms). Drives only the shock section.
- Tool map.** One click jumps to each section; the badge shows its evidence class and the line below says which question the section answers.
- Evidence legend** A–D and the link to this manual.

How valid is a number? — *evidence classes*

The tool separates **what was measured**, **what follows from it by calculation**, **what is an approximation** and **what is an assumption**. Every card carries one of these badges.

CLASS	MEANING	IN PLAIN WORDS	HOW TO USE IT
A • Measured	Measured or digitised from a traceable source case (test report, FE result, DMA datasheet).	A real data point.	Quote it — with its source and conditions.
B • Derived	Dimensionally valid calculation from A-level inputs (e.g. $p = F / (d \cdot L)$).	Plausibly derived — only as good as its inputs.	Use for screening; check the inputs.
C • Heuristic	Calibrated fit or proxy that preserves a directional pattern — not a precise value.	An approximation anchored to one data point.	Read the trend and the direction, not the digits.
D • Hypothesis	Scenario to reveal a possible mechanism — qualitative only.	An estimate based on assumptions and experience.	Use to ask the question — then measure.
A/D • Mixed	Parameter set combining datasheet values (A) with assumed modal values (D).	Partly real, partly assumed.	Check which line is which.

The calibration anchor

The entire model rests on **one calibrated operating point**: power mode • 0° (closed) • Δd 0.200 mm. At this point forces and FE peak stresses are reproduced. Away from it, curve shapes are reconstructed and values extrapolated. The validity pill shows how far you are away:

PARAMETER	CALIBRATED ANCHOR	NEAR ANCHOR	EXTRAPOLATED	FAR EXTRAPOLATION
Clearance Δd	0.200 mm (± 0.0025)	± 0.025 mm	± 0.075 mm	beyond ± 0.075 mm
Guide-vane angle	$\leq 2^\circ$	$\leq 10^\circ$	$> 10^\circ$	—
Operating mode	Power generation	—	Pump (no anchor)	—

The worst of the three parameters sets the overall level. Example: 15° in power mode at 0.200 mm is already *Extrapolated* — because of the angle.

RULE OF THUMB

The closer the pill is to **Calibrated anchor** and the further left the badge is (A → D), the more a number can bear. A **D** result at **Far extrapolation** is a question, not an answer.

Section by section — *what you can read where*

SECTION	CLASS	QUESTION IT ANSWERS	ROBUST TAKEAWAY
Validity window	meta	How far am I from the anchor?	Read before anything else.
01 Bearing forces F1–F3	C	How is the load split between the three bearings?	F2 (middle, nearest the vane) carries most; F3 stays low.
Force vs. angle	C	Where in the closing sweep do forces rise?	Forces ramp in the closing phase ($\sim 28^\circ \rightarrow 18^\circ$) and plateau near closed.
02 System reaction (L2)	C	Which material shows the higher overall load index?	Comparison index only — not a physical resultant.
03 Stress proxy	B / C	How does the local FE peak compare with the limit of the chosen load regime?	bm concentrates stress; tex spreads it. Regime choice flips the limit ranking.
03B Hard-limit screen	B	Does a material fail p, pU or v against its datasheet?	Exclusion test. Capacity rarely decides — mechanisms do.
Vibration spectrum	D	Broad or selective vibration behaviour?	Character only; synthetic, no physical unit.
Resonance position	D	Where does f_0 sit relative to excitation f_a ?	bm's resonance advantage depends on clearance.
Modal parameter set	A/D	Which material parameters drive the dynamics?	High damping (tex) vs. high stiffness (bm).
Shock response	D	When does compliance protect?	Short impulses ($< \sim 10$ ms): tendency tex. Long: bm.
Reports 01 / 02	C	Snapshot / full-cycle summary for sharing	Always carries the validity header — keep it attached.
Scenario I/O	—	Save, share, restore a scenario	Reproducible customer conversations.

ACROSS ALL SECTIONS — NOT MODELLED

Service life, wear rate, years · multi-mode resonances · temperature shift of material properties · squeeze-film damping at tight clearance · misalignment edge-loading · clearance drift from wear · cumulative impact fatigue · cavitation erosion. Nothing in the tool is a turbine simulation, FE solver or CFD model.

OUTPUT 01 + LIVE CHART Bearing forces & force vs. angle

C · HEURISTIC

WHAT YOU SEE

F1 (lower), F2 (middle, nearest the guide vane), F3 (upper) in kN for both materials at the current angle, plus the full closing trajectory. Play draws the curve live like an oscilloscope.

WHAT YOU CAN ASSESS

Load distribution and the phase in which forces rise. In the model tex develops higher absolute forces (compliance lets the shaft deflect), bm stays lower. Δd reacts differently: tex pivots around its optimum (~ 0.15 mm), bm scales more uniformly (optimum ~ 0.12 mm).

WHAT IT DOES NOT TELL YOU

Exact forces away from 0° / 0.200 mm / power mode. Curve shape is reconstructed around one anchor. Use relative statements, not kN values, for anything but the anchor.

OUTPUT 02 System reaction index

C · HEURISTIC

WHAT YOU SEE

$\sqrt{(F1^2 + F2^2 + F3^2)}$ per material and the percentage difference.

WHAT YOU CAN ASSESS

A single comparison number within this model — useful to follow how the gap between the materials changes with angle and Δd .

WHAT IT DOES NOT TELL YOU

It is not a physical force resultant: the reactions act at different bearings.

OUTPUT 03 Stress proxy · load regime

B at anchor

C elsewhere

WHAT YOU SEE

FE-anchored local peak stress per material against the limit of the selected regime (static · dynamic · impact). At the anchor the FE values 54.4 MPa (tex) / 116.1 MPa (bm) are reproduced → class B there, C elsewhere.

WHAT YOU CAN ASSESS

tex distributes load over a broader footprint, bm concentrates it. The regime flips the ranking of the limits: static favours bm (280 vs. 230 MPa), dynamic favours tex (140 vs. 100 MPa). **Select the regime first, then compare.**

WHAT IT DOES NOT TELL YOU

A local peak stress is not projected bearing pressure — utilisation here is indicative. No impact limit is published; the impact setting shows the dynamic limit as a floor and is a test requirement, not a margin.

OUTPUT 03B Hard-limit screen

B · DERIVED · INPUTS C

WHAT YOU SEE

Projected pressure $p = F2 / (d \cdot L)$, sliding speed v and pU per material vs. datasheet limits, with PASS / FAIL. Inputs: bore d , bearing length L , swing β , cycle frequency f .

WHAT YOU CAN ASSESS

A **non-compensatory** exclusion: a material that fails a hard limit is out, whatever its mechanism advantages. If both pass, capacity does not decide — the mechanisms (edge loading, damping, shock) do.

WHAT IT DOES NOT TELL YOU

Geometry is a **placeholder** until you enter the real $d \times L$. $F2$ is model-derived (C away from the anchor). bm limits come from the manual table; bm datasheet not yet in the source pack. No lifetime implied.

LIVE CHART Vibration spectrum

D · HYPOTHESIS

WHAT YOU SEE

Synthetic PSD curves at the three bearing positions, switching with the operating mode; dominant peak at 47 Hz in power mode.

WHAT YOU CAN ASSESS

Character, not magnitude: tex shows broader bandwidth with more peaks; bm a narrower, more selective response — calm unless f_0 meets the excitation. Measured reference ratio tex/bm ≈ 1.58 at the anchor.

WHAT IT DOES NOT TELL YOU

Spectra are illustrative, have no physical unit and are not derived from $|H(f)|$. Aggregated over the closing operation, not instantaneous.

LIVE CHART Resonance position

D · HYPOTHESIS

WHAT YOU SEE

Natural frequency f_0 of each material vs. dominant excitation f_a (47 Hz power mode), with zones: resonance (± 7 Hz), transition, safe ($> 2 \cdot f_a$).

WHAT YOU CAN ASSESS

At the anchor tex sits close to f_a (~ 49 Hz) but is heavily damped ($Q \approx 3.8$); bm sits higher (~ 62 Hz) with a sharp $Q \approx 50$. bm's advantage is **clearance-conditional**: bm's f_0 drops about twice as fast with Δd (-120 vs. -60 Hz/mm) and can fall into the resonance zone at wide clearance.

WHAT IT DOES NOT TELL YOU

f_0 is assumed — back-calculated from one PSD ratio, which cannot uniquely identify two natural frequencies. No resonance prediction; only a modal impact test on the installed system can tell.

REFERENCE Modal parameter set

A/D · MIXED

WHAT YOU SEE

E' , $\tan \delta$, ζ , Q , f_0 , $|H(f_a)|$ and friction coefficient per material (DMA reference at 50 Hz, room temperature).

WHAT YOU CAN ASSESS

The physical reason behind all dynamic sections: tex = low stiffness ($E' \approx 3.5$ GPa), high damping ($\tan \delta \approx 0.26$); bm = high stiffness (≈ 105 GPa), low damping (≈ 0.02). Everything dynamic in the tool follows from this contrast.

WHAT IT DOES NOT TELL YOU

Which line is measured and which assumed must be read per value: modulus from datasheet (A), f_0 and damping fitted (D). No temperature dependence.

LIVE CHART Shock & impact response

D · HYPOTHESIS

WHAT YOU SEE

Normalised force response to a pulse of duration τ (1 ms cavitation ... 50 ms quasi-static), optional shock patterns during the sweep (cavitation, vortex, water hammer, mixed; density, intensity, seed).

WHAT YOU CAN ASSESS

Where the verdict swaps: for short impulses ($< \sim 10$ ms) compliance becomes protective — tendency toward tex (peak reduction in the order of one third); for long pulses stiffness dominates — tendency toward bm. This is the section where "elastic or rigid" is decided by the event spectrum.

WHAT IT DOES NOT TELL YOU

Single-degree-of-freedom, first-order filter ($\tau_{\text{char}} \approx 5$ ms) — qualitative only. No fatigue accumulation. Requires a site-specific shock spectrum to become an argument.

REPORTS · PERSISTENCE Snapshot, cycle report, scenario I/O

C · HEURISTIC

WHAT YOU SEE

Report 01 captures the current point, Report 02 the full $90^\circ \rightarrow 0^\circ$ sweep with dominant failure-mode assessment. Both export as PDF / text. Scenario I/O saves state + full sweep (1° steps) as JSON.

WHAT YOU CAN ASSESS

A shareable, reproducible record of a conversation — which point was discussed, under which assumptions. The sweep export can be analysed in Excel or Python.

WHAT IT DOES NOT TELL YOU

Reports combine C- and D-class modules. Every report carries a validity header — never forward a report without it.

Using it in a conversation — *from screen to next question*

- 1 **Start at the anchor.** Power mode, 0°, 0.200 mm. Show what is measured and what the tool reproduces.
- 2 **Enter the real geometry** in the hard-limit screen. If a material fails, it is out — the conversation is about the other one.
- 3 **Select the load regime** that matches the customer's situation (static, alternating, impact) before comparing stress against limits.
- 4 **Move one slider at a time** and watch the validity pill. Name the evidence class aloud when you quote a number.
- 5 **Find the mechanism.** Edge loading? Resonance near f_a ? Frequent short impulses? The mechanism, not the material, is the result.
- 6 **End with the open question** and generate the report as the record of what was discussed.

TYPICAL NEXT QUESTIONS THE TOOL LEADS TO

- What is the real bearing geometry $d \times L$ and the clearance definition (radial / diametral)?
- What is the actual excitation spectrum — is there a dominant peak near 47 Hz?
- Are there frequent short events: cavitation, debris, vortex bursts, water hammer?
- Is misalignment or edge loading known from inspection?
- Can a modal impact test of the installed system be arranged?
- What is the clearance today and how will it drift with wear?

SENTENCES THAT STAY TRUE

"The tool shows a tendency under stated assumptions." · "This value is anchored / extrapolated / assumed." · "To decide this we need a measurement."

SENTENCES TO AVOID

"The tool proves..." · "Material X lasts N years." · "The force at 30° is exactly ... kN." · "There is no resonance risk."

Source basis

Calibration anchor from a manufacturer test report on these compound classes (Pereira, HYDRO 2025 — source pack pending): FE contact pressure at closed position (54.4 MPa tex / 116.1 MPa bm), PSD spectra at Δd 0.200 mm (47 Hz dominant peak, ratio tex/bm 1.58), bearing forces F1–F3 across angle. Material limits: tex 552 W87.00 bushing column; bm 392 per DEVA manual table 3.1.B. DMA reference data at 50 Hz, room temperature. All other values are reconstructed, extrapolated or assumed as labelled in the tool.